HUMAN RESOURCES ANALYSIS DASHBOARD

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ROLE :- BUSINESS ANALYST PROJECT

*PROJECT OVERVIEW :-

- **Human Resources Analysis Dashboard was developed to analyze workforce data and support HR decision-making using visual insights.**
- **The project focuses on key HR metrics such as employee demographics, hiring trends, employee turnover, performance metrics, and employee satisfaction.**
- **Various data visualization techniques including bar charts, line charts, histograms, and pie charts were used to present HR insights clearly.**
- **The dashboard helps identify hiring patterns, workforce distribution, employee performance trends, and turnover reasons.**
- **This analysis enables HR teams to improve recruitment strategies, enhance employee retention, and monitor workforce satisfaction effectively.**

*DATASET OVERVIEW :-

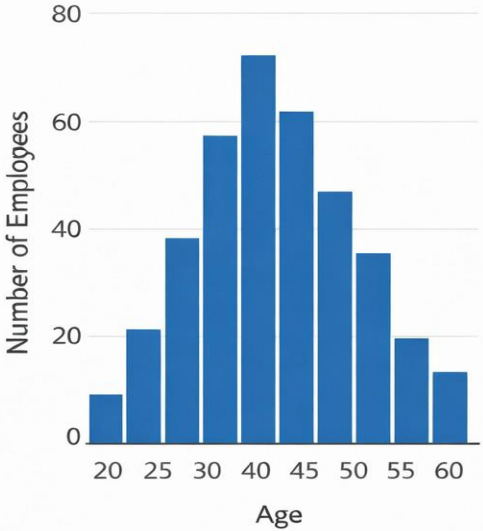
- The dataset contains 311 employee records including demographic, employment, and performance information.
- It includes attributes such as age, gender, department, recruitment source, hire date, termination details, and employee satisfaction scores.
- The dataset provides information about employee turnover, termination reasons, and hiring sources.
- It also contains performance ratings and satisfaction levels which help evaluate workforce engagement and productivity.
- The dataset is useful for analyzing workforce structure, recruitment effectiveness, employee retention, and overall HR performance.

*KPI CARDS :-

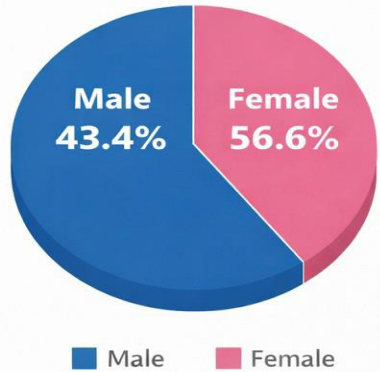
KPI'S	Values
Total Employees	311
Turnover Rate	33.4 %
Average Performance Score	N/A (categorical)
Average Satisfaction Score	3.89 / 5

Employee Demographics

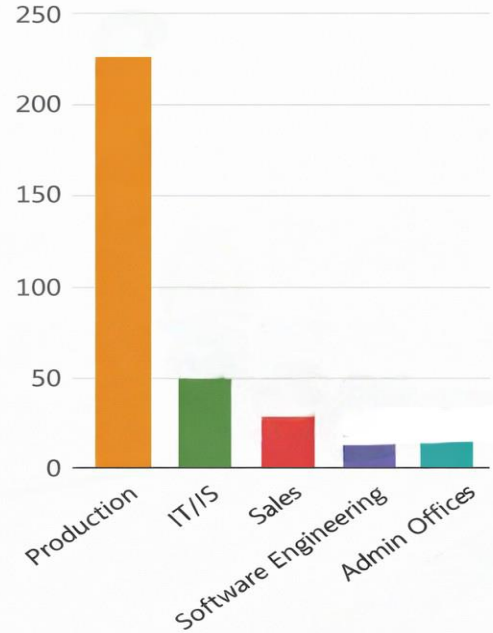
Age Distribution



Gender Distribution



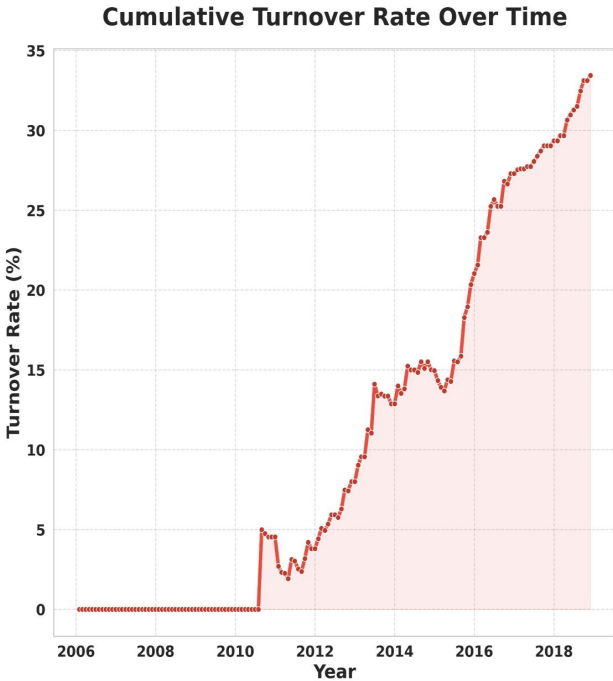
Department Distribution

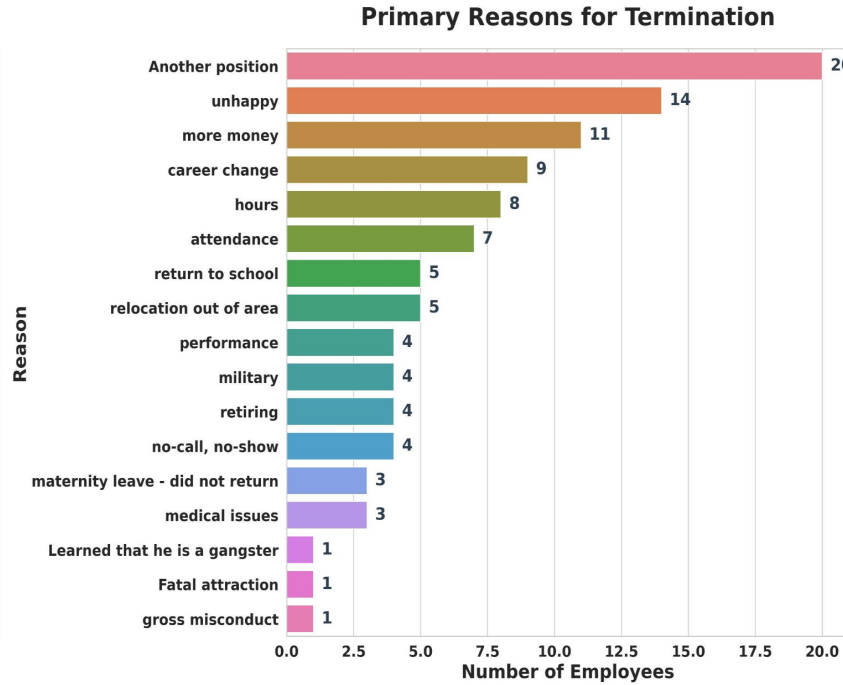


*INSIGHTS FROM EMPLOYEE DEMOGRAPHICS :-

- **Heavy Operational Lean:** The organization is overwhelmingly dominated by the Production department (over 200 employees), making the company highly sensitive to labor trends and turnover within that specific sector.
- **Mature Workforce:** The age distribution peaks between 35 and 45, suggesting an experienced workforce. However, the company may face a "knowledge gap" in the next 10–15 years as this large cohort nears retirement.
- **Gender Balance:** The workforce is relatively balanced but leans female (56.6%). Management should ensure that internal policies (like the "maternity leave" turnover mentioned in the other chart) are optimized to retain this majority.
- **Scaling Bottleneck:** There is a massive disparity between Production and support roles like IT/IS and Software Engineering. If the company intends to modernize or automate, the current technical staff levels may be insufficient to support the Production volume.
- **Recruitment Focus:** The low number of employees in the 20–25 age bracket indicates a potential struggle to attract "early career" talent or Gen Z workers, which could impact long-term sustainability.

Employee Turnover Analysis

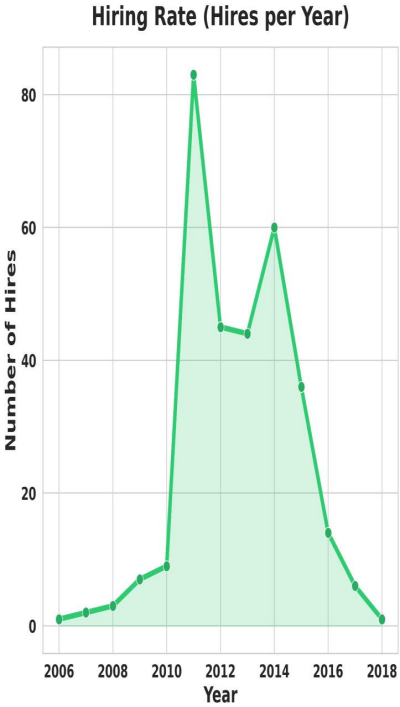


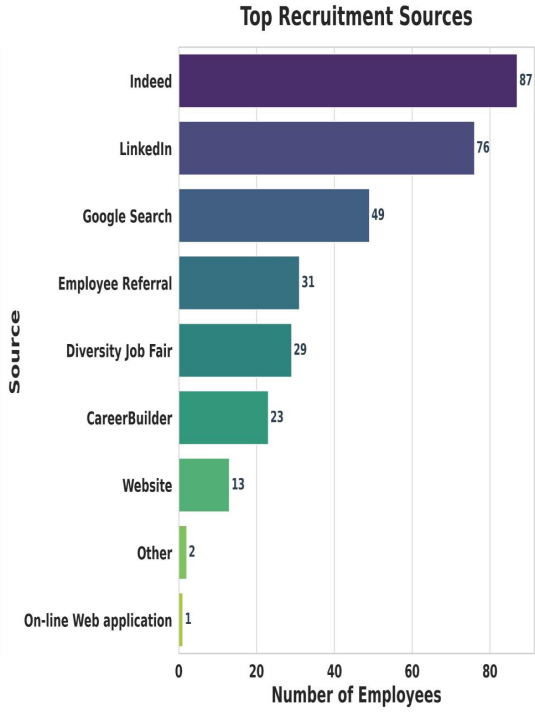


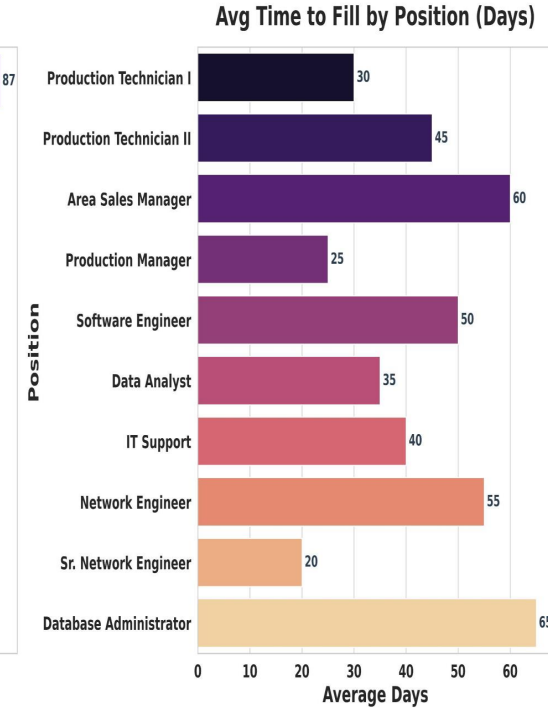
*INSIGHTS FROM EMPLOYEE TURNOVER ANALYSIS :-

- **Accelerating Turnover Trend: After remaining stagnant until 2010, the cumulative turnover rate has climbed aggressively, reaching over 33% by 2019. This suggests a systemic issue that has worsened over the last decade.**
- **Dominance of Voluntary Resignation: The top three reasons—"Another position" (20), "Unhappy" (14), and "More money" (11)—indicate that 45% of terminations are likely preventable through better retention strategies or competitive pay.**
- **Cultural Red Flags: "Unhappy" is the second leading cause of exit. This points to a potential issue with company culture, management styles, or work-life balance that needs immediate internal investigation.**
- **Attendance & Discipline: Performance-related issues like "Attendance" and "No-call, no-show" are significant contributors, suggesting a need for more robust screening during the hiring process or clearer workplace policies.**
- **Statistical Outliers: The data includes rare, high-risk termination reasons (e.g., "Learned that he is a gangster," "Fatal attraction"). While statistically low, they highlight a need for stricter background checks or security protocols.**

Hiring Trends Analysis



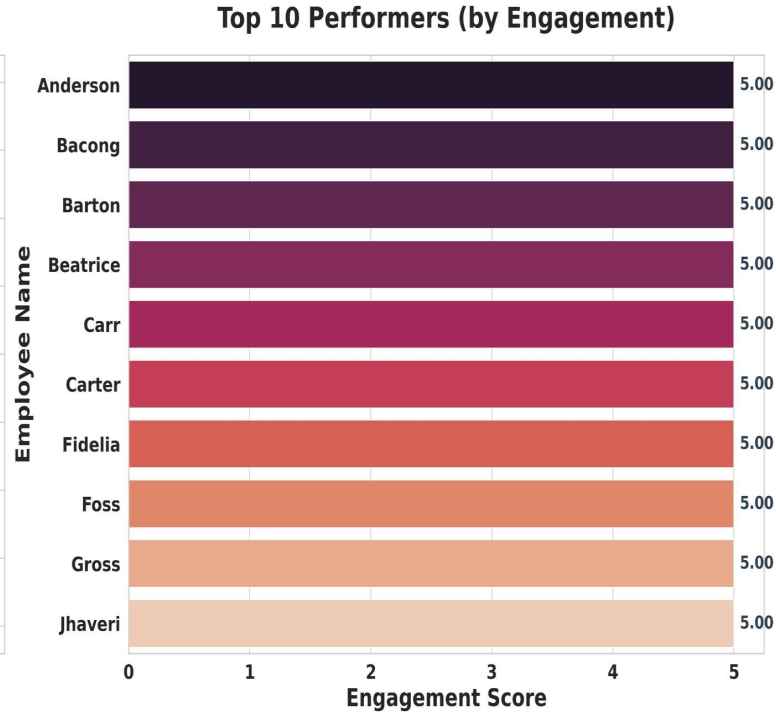




*INSIGHTS FROM HIRING TRENDS ANALYSIS :-

- **Massive Hiring Peak and Sharp Decline: Hiring reached a major peak in 2011 (over 80 hires), followed by a secondary peak in 2014, but has crashed to near-zero as of 2018.**
- **Digital Sourcing Dominance: Indeed (87) and LinkedIn (76) are the primary drivers of successful hires, significantly outperforming employee referrals and job fairs.**
- **Technical Role Difficulty: Specialized roles like Database Administrator (65 days) and Area Sales Manager (60 days) take the longest to fill, indicating a tight market for these skill sets.**
- **Recruitment Efficiency Gap: While Sr. Network Engineers are filled quickly (20 days), general Network Engineers take nearly triple the time (55 days), suggesting an imbalance in the candidate pipeline for mid-level roles.**
- **Referral Underutilization: Despite being a high-quality source in many industries, Employee Referrals (31) represent less than half the volume of Indeed, suggesting an untapped opportunity for a referral incentive program.**

Performance Metrics Analysis

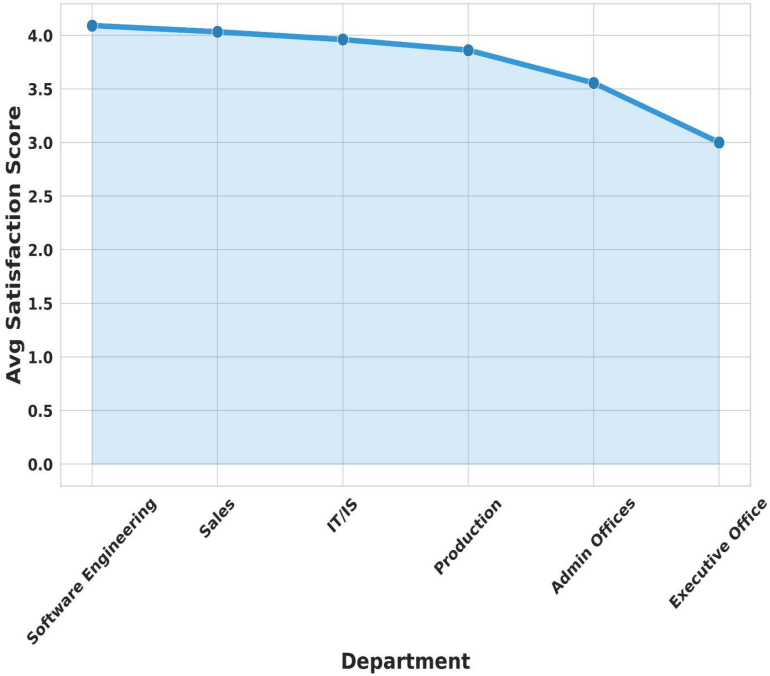


*INSIGHTS FROM PERFORMANCE METRICS ANALYSIS :-

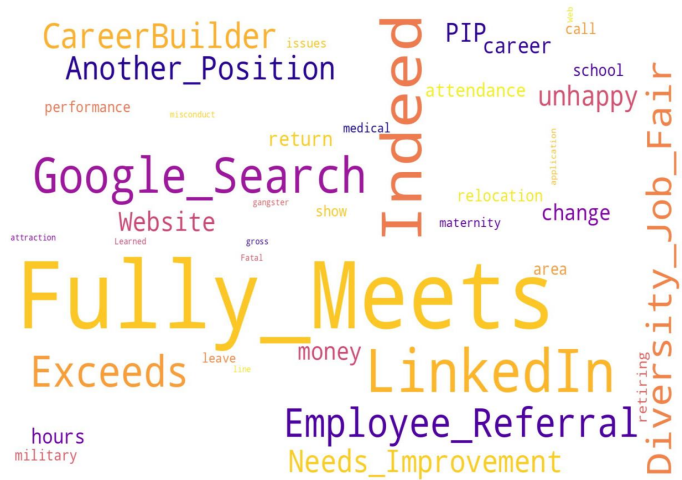
- **High Engagement Correlates with Performance:** The scatter plot shows a dense cluster of "Exceeds" and "Fully Meets" performance scores at engagement levels above 3.0.
- **Satisfaction/Engagement Disconnect:** Several employees on PIPs (Performance Improvement Plans) actually report moderate satisfaction (scores of 2.0 to 3.0), suggesting that some low performers may be content but unproductive.
- **Top Talent Benchmarks:** The top 10 performers all share a perfect 5.00 engagement score, highlighting that high engagement is a primary indicator of top-tier individual performance.
- **Vulnerability in High-Satisfaction Low-Performers:** There are scattered "Needs Improvement" data points with high satisfaction scores (4.0–5.0), which could indicate a "comfort zone" where lack of challenge leads to underperformance.
- **Critical Zone:** Employees with engagement and satisfaction scores below 2.0 are almost exclusively categorized as "PIP" or "Needs Improvement," marking a clear danger zone for retention and productivity.

Employee Satisfaction & Feedback Analysis

Average Satisfaction by Department



Employee Feedback Word Cloud (Dataset-Driven)



*INSIGHTS FROM EMPLOYEE SATISFACTION AND FEEDBACK ANALYSIS :-

- **Departmental Disparity: Software Engineering reports the highest average satisfaction (above 4.0), while the Executive Office reports the lowest (3.0), suggesting a possible disconnect or higher stress at the leadership level.**
- **Production Stability: The Production department maintains a solid satisfaction score of nearly 4.0, which is positive given it is the largest department by headcount.**
- **Feedback Sentiment: The word cloud indicates that "Fully Meets" and "Exceeds" are the most dominant themes in employee feedback, signaling a generally positive performance culture.**
- **Turnover Drivers: Words like "Another_Position," "unhappy," and "money" appear prominently in the feedback data, confirming that competitive pay and career growth are the primary threats to retention.**
- **Recruitment Branding: The heavy presence of "Indeed," "Google_Search," and "LinkedIn" in the feedback word cloud suggests that employees strongly associate their journey with the organization starting from these specific external platforms.**

*CONCLUSION :-

- The Human Resources Analysis Dashboard provides clear insights into employee demographics, hiring trends, turnover patterns, performance levels, and employee satisfaction.
- The analysis shows that understanding workforce distribution and hiring sources helps organizations improve their recruitment strategies.
- Employee turnover trends and termination reasons highlight the importance of employee retention strategies and workplace improvements.
- Performance and satisfaction metrics help HR teams monitor employee engagement and overall workforce productivity.
- Overall, the dashboard supports data-driven HR decision making, helping organizations optimize hiring, improve employee satisfaction, and maintain a stable workforce.

Human Resources Analysis Dashboard

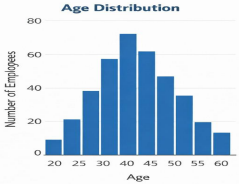
Total Employees
311

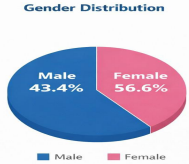
Turnover Rate
33.4%

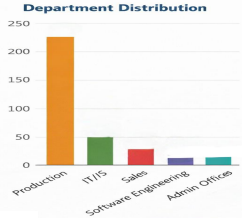
Avg Performance Score
N/A (categorical)

Avg Satisfaction Score
3.89

Employee Demographics







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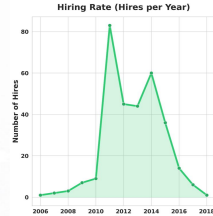

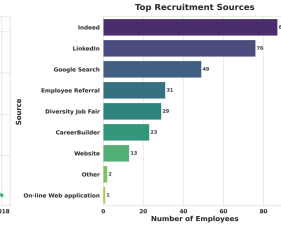

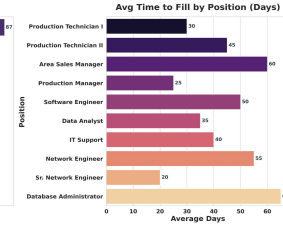

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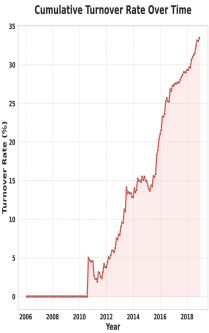
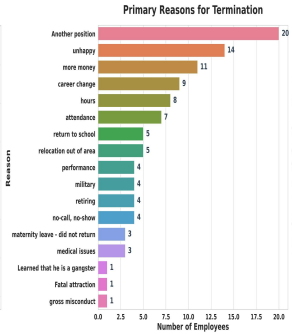
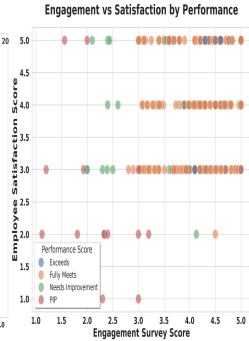
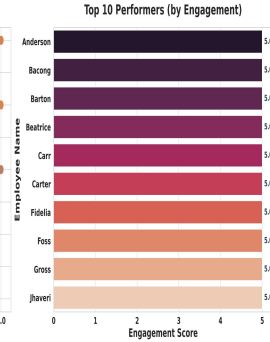
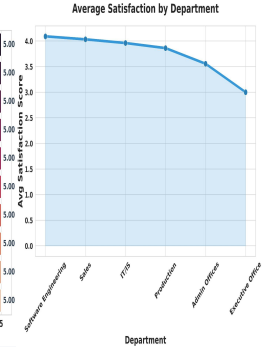




Performance Metrics Analysis







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